

Lecture 5: Joint and Marginal Distributions, Conditioning, Independence

Jamie Haddock

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1 Why look at two random variables together?

So far we've studied one random variable at a time. But often we care about how *two* random variables move together — two dice rolled at once, or (in randomized Kaczmarz) the row index sampled at one iteration versus the next. Knowing each RV's own PMF separately says nothing about how they *relate*.

i Definition: Joint PMF

For discrete random variables X, Y on the same probability space,

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

Like any PMF, it's nonnegative and sums to 1 over all pairs (x, y) . It's strictly more informative than p_X and p_Y separately — it tells us how X and Y *co-vary*.

Two-die-roll example: Let X = first die, Y = second die – all 36 outcomes $(x, y) \in \{1, \dots, 6\}^2$ are equally likely, so

$$p_{X,Y}(x, y) = \frac{1}{36}, \quad x, y \in \{1, \dots, 6\}.$$

A Joint distributions of arbitrarily many RVs!

We may consider joint PMFs of any number of RVs defined on the same probability space, $p_{\{X_i\}_{i=1}^k}(\{x_i\}_{i=1}^k) = P(X_1 = x_1, \dots, X_k = x_k)$.

2 Marginal PMF

i Definition: Marginal PMF

Given a joint PMF $p_{X,Y}$, the **marginal PMF** of X is recovered by summing out y :

$$p_X(x) = P(X = x) = \sum_y p_{X,Y}(x, y),$$

and symmetrically $p_Y(y) = \sum_x p_{X,Y}(x, y)$.

Two-die-roll example: summing each row of the 36-entry joint table over y gives

$$p_X(x) = \sum_{y=1}^6 \frac{1}{36} = \frac{6}{36} = \frac{1}{6}, \quad x = 1, \dots, 6,$$

recovering the familiar single-die PMF.

3 Expectation of a function of two RVs

i Theorem: LOTUS for two random variables

For random variables X, Y and any function g ,

$$E[g(X, Y)] = \sum_{x,y} g(x, y) p_{X,Y}(x, y).$$

Taking $g(x, y) = ax + by + c$ recovers linearity of expectation that we saw previously:

$$\begin{aligned} E[aX + bY + c] &= \sum_{x,y} (ax + by + c) p_{X,Y}(x, y) = a \sum_x x p_X(x) + b \sum_y y p_Y(y) + c \\ &= aE[X] + bE[Y] + c. \end{aligned}$$

(The middle step uses the marginal-PMF definition — summing y out of $ax p_{X,Y}(x, y)$ leaves $ax p_X(x)$, and likewise for the y term.)

4 Example: the hat-check problem

n people check their hats at a party; the hats are returned to them in a uniformly random order (a uniformly random permutation of the n hats). Let X = number of people who get their own hat back. Find $E[X]$.

Write $X = \sum_{i=1}^n X_i$, where $X_i = \mathbf{1}_{\{\text{person } i \text{ gets their own hat back}\}}$. By symmetry every person is equally likely to receive any of the n hats, so $P(X_i = 1) = \frac{1}{n}$ and $E[X_i] = \frac{1}{n}$.

 The X_i are *not* independent!

If the first $n - 1$ people all get their own hat back, the last person is guaranteed to as well — knowing some X_i 's tells you a lot about the others. But, just like the expectation of the Binomial calculation we saw previously, **linearity doesn't care**:

$$E[X] = \sum_{i=1}^n E[X_i] = n \cdot \frac{1}{n} = 1.$$

However many people are at the party, on average exactly **one** person gets their own hat back!

5 Conditioning

We've used conditional probability previously — let's now define *conditional distributions* of random variables precisely.

 Definition: Conditional PMF given an event

For an event A with $P(A) > 0$,

$$p_{X|A}(x) = P(X = x \mid A) = \frac{P(\{X = x\} \cap A)}{P(A)}.$$

Revisiting our die-roll example: let X be a single fair die roll and $A = \{X \text{ is even}\} = \{2, 4, 6\}$. Then

$$p_{X|A}(x) = \begin{cases} \frac{1}{3} & x \in \{2, 4, 6\} \\ 0 & \text{otherwise,} \end{cases}.$$

6 Conditioning on another random variable

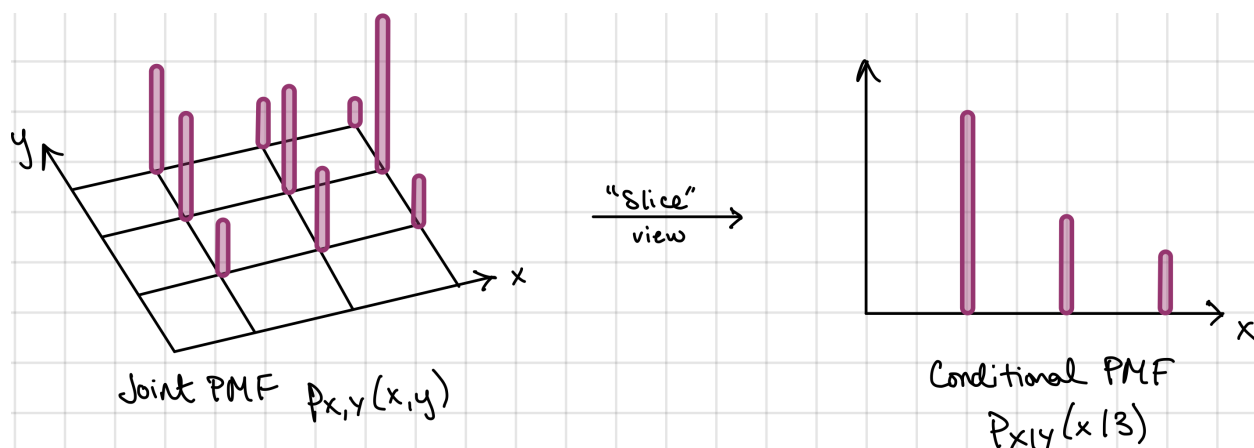
 Definition: Conditional PMF given $Y = y$

For $p_Y(y) > 0$,

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

This is exactly the previous definition applied to the event $A = \{Y = y\}$.

7 Grid view of the joint PMF, slice view of the conditional PMF



8 Recovering the marginal from conditional PMFs

i Theorem (total probability, for random variables)

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y).$$

This is just $p_X(x) = \sum_y p_{X,Y}(x,y)$ (the marginal definition) with $p_{X,Y}(x,y) = p_Y(y) p_{X|Y}(x|y)$ substituted in.

9 Conditional expectation

i Definitions: conditional expectation

$$E[X | A] = \sum_x x p_{X|A}(x), \quad E[X | Y = y] = \sum_x x p_{X|Y}(x|y).$$

Same idea as ordinary expectation — just averaging x against the *conditional* PMF instead of the unconditional one.

10 Law of total expectation – three forms

- **Partition form:** if $A_1, \dots, A_n$ partition the sample space,

$$E[X] = \sum_{i=1}^n P(A_i) E[X | A_i].$$

- **Conditioned partition form:** for any event B ,

$$E[X | B] = \sum_{i=1}^n P(A_i | B) E[X | A_i \cap B].$$

- **Random-variable form:**

$$E[X] = \sum_y p_Y(y) E[X | Y = y].$$

All three say the same thing: split the world into (disjoint) pieces, compute the (conditional) expectation on each piece, and average the pieces back together weighted by how likely each one is.

This is a useful computational trick: whenever a direct computation of $E[X]$ looks hard, look for a natural way to split on “what happened first,” compute the easy conditional pieces, and average.

11 Example: mean and variance of the Geometric, the slick way

Let $X \sim \text{Geometric}(p)$ (number of trials until the first success). Condition on the first trial: $A = \{\text{trial 1 succeeds}\}$, $P(A) = p$.

- If A occurs, $X = 1$, so $E[X | A] = 1$.
- If A^c occurs, that trial was wasted and — by memorylessness — we’re back to square one: $X = 1 + X'$ for a fresh $\text{Geometric}(p)$ copy X' , so $E[X | A^c] = 1 + E[X]$.

By the law of total expectation,

$$\begin{aligned} E[X] &= P(A)E[X | A] + P(A^c)E[X | A^c] = p \cdot 1 + (1 - p)(1 + E[X]) \\ &= 1 + (1 - p)E[X], \end{aligned}$$

so $p E[X] = 1$, giving $E[X] = \frac{1}{p}$ — which matches our previous calculation.

The same trick on X^2 : $X^2 | A^c = (1 + X')^2 = 1 + 2X' + X'^2$, so

$$E[X^2] = p + (1 - p)(1 + 2E[X] + E[X^2]) = 1 + 2(1 - p)E[X] + (1 - p)E[X^2],$$

which solves to $E[X^2] = \frac{2 - p}{p^2}$, and therefore $\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1 - p}{p^2}$ — which is what we saw in a previous lecture!

12 Independence

To define independence for random variables, we build on our previous definition for events.

i Definition: independence of a random variable from an event

X is independent of event A if

$$p_{X|A}(x) = p_X(x) \quad \text{for every } x,$$

i.e. learning that A occurred tells you nothing new about X . Equivalently, $P(X = x, A) = P(X = x)P(A)$ for every x — the L2 definition of independent events, applied to $\{X = x\}$ and A .

13 Independence of two random variables

i Definition: independence of random variables

X and Y are **independent** if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y,$$

equivalently $p_{X|Y}(x | y) = p_X(x)$ whenever $p_Y(y) > 0$ — conditioning on Y never changes X ’s distribution.

14 Conditional Independence

i Definition: conditional independence

X and Y are **conditionally independent given event C** if

$$p_{X,Y|C}(x,y) = p_{X|C}(x)p_{Y|C}(y) \quad \text{for all } x,y.$$

⚠ Recall: these are genuinely different notions!

A pair of random variables can be independent but not conditionally independent given some C , or conditionally independent given C without being (unconditionally) independent — exactly like the two-coin examples from our previous independence discussion.

15 A key fact: $E[XY] = E[X]E[Y]$ for independent X, Y

i Fact

If X, Y are independent, $E[XY] = E[X]E[Y]$.

Proof: apply two-variable LOTUS with $g(x,y) = xy$, then use independence:

$$\begin{aligned} E[XY] &= \sum_{x,y} xy p_{X,Y}(x,y) = \sum_{x,y} xy p_X(x)p_Y(y) = \left(\sum_x x p_X(x) \right) \left(\sum_y y p_Y(y) \right) \\ &= E[X]E[Y]. \quad \blacksquare \end{aligned}$$

⚠ Independence is essential here

Contrast with linearity of expectation, which needed nothing. In general $E[XY] \neq E[X]E[Y]$: e.g. take $Y = X$, so $E[XY] = E[X^2] \neq (E[X])^2 = E[X]E[Y]$ unless $\text{Var}(X) = 0$.

16 Variance of a sum (with proof!)

i Theorem (stated previously): for independent X, Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof: let $\mu_X = E[X], \mu_Y = E[Y]$. Expanding and using linearity twice,

$$\begin{aligned} \text{Var}(X + Y) &= E[((X - \mu_X) + (Y - \mu_Y))^2] \\ &= \text{Var}(X) + \text{Var}(Y) + 2 E[(X - \mu_X)(Y - \mu_Y)]. \end{aligned}$$

Expanding the cross term and applying the fact above,

$$E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y = E[X]E[Y] - E[X]E[Y] = 0.$$

⚠ The cross term

That cross term, $E[(X - \mu_X)(Y - \mu_Y)]$, is called the **covariance** of X, Y . It vanishes exactly because of independence; in general $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

17 Example: mean and variance of the sample mean

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 , and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

$$E[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \mu \quad (\text{linearity — independence not even needed}).$$

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n} \quad (\text{independence used here!})$$

The sample mean is unbiased, and its spread shrinks like $1/n$ — the mathematical reason averaging more samples reduces noise, and the seed of the Law of Large Numbers (coming later this course).

```
import matplotlib.pyplot as plt

rng = np.random.default_rng(151)
n_values = np.array([1, 2, 5, 10, 20, 50, 100, 200, 500])
repetitions = 20_000
sigma_sq = np.var(np.arange(1, 7)) # variance of a single fair die roll

empirical_var = []
for n in n_values:
    sample_means = rng.integers(1, 7, size=(repetitions, n)).mean(axis=1)
    empirical_var.append(sample_means.var())
empirical_var = np.array(empirical_var)

fig, ax = plt.subplots(figsize=(5.5, 3.5))
ax.loglog(n_values, empirical_var, 'o', label=r'empirical $\mathrm{Var}(\bar{X}_n)$')
ax.loglog(n_values, sigma_sq / n_values, '--', label=r'$\sigma^2/n$')
ax.set_xlabel('$n$')
ax.set_ylabel('variance')
ax.legend()
plt.tight_layout()
plt.show()
```

